Rising biocapacity total can mask ecological-capital land drawdown: a two-land model

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Abstract

Biocapacity is reported as an aggregate, although land classes differ in yield and regeneration. A rising total can conceal a change in composition, but it cannot by itself establish that a physical stock has declined. We examine this accounting problem with a two-land model that separates fast-provisioning land from ecological-capital land and allows delays in regeneration and population response. When a shortfall triggers a transfer to fast land, total model biocapacity rises if fast land contributes more per hectare, even as capital-land area falls.

In a representative run, capital-land area falls 46.16% and its modeled contribution falls 51.72%, while total model biocapacity rises 39%. A separate fixed-demand comparison reaches its minimum-area boundary at a demand ceiling of about 1.14 gha. The reduced branch has no fold, hysteresis, or critical slowing; this does not characterize every trajectory of the delayed model. In NFA Edition 2026 (1961–2023), total world biocapacity account value rises 24.86%, while B/P falls 52.71% and the production footprint per person rises 16.73%. Cropland contributes 24.06 percentage points, Forest biocapacity subtracts 5.95 points, and built-up land adds 7.44 as an accounting class. These account movements are not physical-stock measurements. The results show why component values should accompany aggregate totals.

Keywords: biocapacity, ecological footprint, land-use conversion, aggregation, identifiability, critical slowing down, delay differential equations, carrying capacity

1. Introduction

Carrying capacity is the population a resource base can sustain. The resource base's condition and composition may change with use, yet models often treat capacity as fixed because it is difficult to estimate (Cohen, 1995).

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We ask a narrower accounting question: can total biocapacity rise while a productive land class loses area, and what can the aggregate reveal about that change?

The National Footprint and Biocapacity Accounts (NFA) report biocapacity as annual biological supply and the Ecological Footprint as annual demand on biologically productive area. Both are expressed in global hectares (gha), which adjust physical area by biological productivity relative to the world average for the accounting year; the annual basis is implicit (Global Footprint Network, 2009, 2019). Because an account total combines area and productivity, its change alone cannot show whether yield, area, or the mix of land classes changed. Composite-indicator research describes this as *compensation*: a gain in one component can offset a loss in another (Nardo et al., 2008; Becker et al., 2017; Fischer et al., 2022). Related work examines how the Ecological Footprint should be interpreted (Blomqvist et al., 2013; Giampietro and Saltelli, 2014; van den Bergh and Grazi, 2015; Galli et al., 2016). The aggregate can be informative, but it is more interpretable beside its component values.

Human–environment models often summarize nature as a single stock or overall balance. Examples include the World3 system (Meadows et al., 1972, 1974, 2004), models of Easter Island (Brander and Taylor, 1998; Anderies, 2000), and coupled Earth-system frameworks (Motesharrei et al., 2016; Donges et al., 2017; Calvin and Bond-Lamberty, 2018; Henderson and Loreau, 2018). Other work asks whether carrying capacity is fixed, variable, or useful as a concept (Cohen, 1995; van den Bergh and Rietveld, 2004; Simon, 1996; Meyer, 1994; Meyer and Ausubel, 1999; Dasgupta, 2021); global demographic assessments often project population separately from environmental constraints (United Nations, 2022). Land-use studies document conversion among forests, grasslands, wetlands, and cropland (Haberl et al., 2007; Krausmann et al., 2013; Erb et al., 2017). These literatures motivate examining land classes with different yields and regeneration times. They do not make NFA account trends evidence of a particular land conversion or physical-stock change.

The orchard and the elevator. Imagine an orchard with fast-provisioning ground, such as intensively managed beds, beside slower-renewing ground that supports trees and soil. The annual harvest is a flow; the assets that support future harvests are stocks.

In the model, a shortfall can trigger a transfer from ecological-capital land to fast-provisioning land. If the receiving class contributes more per hectare in that scenario, total model biocapacity can rise while capital-land area falls. This illustrates a conditional composition mechanism; it is not evidence that NFA values measure physical stocks or that the accounts document such a transfer.

An elevator offers a second analogy. A car rated for ten passengers might continue to run with fourteen, but the ride alone does not reveal the condition of its cables. Service and asset condition are different quantities. The analogy does not predict that failure is inevitable or sudden; it illustrates why observing a delivered flow cannot by itself identify the state of the asset that supports it.

We ask three questions. First, under what model conditions can conversion raise total biocapacity while A_c falls, and how long can that divergence last? Second, what does the reduced fixed-demand comparison imply about its balance branch, and what remains path-dependent in the delayed system? Third, what limits drawdown in the model, and what explicit flow permits recovery?

The paper makes three contributions. It develops a two-land model that conserves total area, tracks capital-land quality separately, and includes delayed regeneration and population response. It derives the condition under which conversion contributes positively to total model biocapacity and distinguishes the reduced fixed-demand balance branch from trajectories of the full delayed system. It also decomposes NFA Edition 2026 account changes and examines an auxiliary crop-area/yield proxy while stating what these account values can and cannot identify. Five testable implications close the analysis. Parameters are representative rather than fitted, so the simulations illustrate conditional mechanisms rather than historical forecasts.

Related material-ledger work describes a broader flow–stock masking argument as a *productivity illusion*; here we examine its composition-specific form (Abaee, 2026c).[1] The ledger comparison is an analogy about compensation, not an identity between biocapacity and material throughput.

[1]The v50 PDF is present at the pinned commit. The checksum manifest lists the v49 article files but omits v50; the README dated 2026-09-19 does not explicitly label v49 or v50; and the v50 TeX header points to the v48 Markdown source.

2. Methods

2.1. A two-land model with conserved area and ecosystem quality

We divide a fixed total bioproductive area A_{tot} among three mutually exclusive model classes: fast-provisioning land A_f, ecological-capital land A_c, and reserve land A_r. Fast land represents cropland and other intensively managed systems; capital land groups forest, soil, wetland, and long-rotation systems into one effective class, not one real land type. Reserve land includes fallow, retired, or not-yet-allocated area. The class areas sum to A_{tot}.

Capital land has a normalized quality state $q \in [0, q_{\max}]$. This effective state changes its account value per hectare; it is not a direct measurement of biomass, soil integrity, or ecosystem health. In the model, quality follows a logistic growth rule:

$$G_c(q) = \rho_c \, q \left(1 - \frac{q}{q_{\max}}\right), \tag{1}$$

The annual account contribution per hectare of capital land combines a direct-yield term with a term linked to quality growth:

$$\Upsilon_c(q) = b_c + b_{G,c} \, G_c(q), \tag{2}$$

Fast land contributes $Y_f = b_f A_f$, while capital land contributes $Y_c = \Upsilon_c(q) A_c$. Their sum is total model biocapacity, an annual account-equivalent amount in gha with the annual basis implicit (Global Footprint Network, 2009, 2019):

$$B = Y_f + Y_c = b_f A_f + \Upsilon_c(q) \, A_c, \tag{3}$$

Population P determines demand through the constant per-person footprint e, so $E = eP$. The shortfall S is the part of demand not met by the human-available shares of the two modeled contributions:

$$S = \left[E - \sigma_f Y_f - \sigma_c Y_c\right]_+, \tag{4}$$

The positive-part operator is $[x]_+ = \max(x, 0)$. In the baseline, both shares equal one, so $S = [E - B]_+$; lower shares are a reservation extension. When a shortfall remains, the model transfers capital-land area to fast land at a finite rate:

$$u_c = \frac{1}{\tau_{\text{conv}}} \min\left(\frac{S}{b_f},\ A_c - A_c^{\min}\right) \qquad [\text{ha}\,\text{yr}^{-1}], \tag{5}$$

Here S/b_f is the fast-land area needed to cover the current shortfall, $A_c^{\min}$ is the typed capital-land floor, and τ_{conv} sets the conversion timescale. This is the model's transfer rule; it is not an empirical reconstruction of clearing behavior. Land moves among the three classes through one-for-one transfers:

$$\dot{A}_c = -u_c + R_{rc}, \tag{6}$$

$$\dot{A}_f = u_c + \mu A_r - \eta_f A_f, \tag{7}$$

$$\dot{A}_r = -\mu A_r + \eta_f A_f - R_{rc}, \tag{8}$$

The flow μA_r moves reserve land into fast use, while $\eta_f A_f$ retires fast land to reserve. Reserve land returns to capital land only through the explicit surplus-gated restoration flow:

$$R_{rc} = \chi_r\, A_r\, [B - E]_+\,, \tag{9}$$

These transfers conserve total area, with A_r determined by the area balance. The model has no direct fast-land-to-capital-land return route. Without restoration, conversion stops when the shortfall closes, but a lower population or a closed shortfall does not by itself restore the capital-land area. Reserve replanting can maintain fast land when conversion is inactive; without it, retirement eventually reduces fast-land area to zero at a no-conversion equilibrium.

Quality and population respond with delays. Quality follows its delayed growth rule, and population follows a delayed logistic law (Pearl, 1925; Hutchinson, 1948); carrying capacity is based on modeled biocapacity per person and has a typed floor. Shortfalls also drive a leaky debt state that reduces fast-land yield:

$$\dot{q} = G_c\big(q(t-\tau_g)\big), \tag{10}$$

$$\dot{P} = rP\left(1 - \frac{P}{K(t-\tau_p)}\right), \qquad K = \max(B/e,\, K_{\min}), \tag{11}$$

$$b_f = \big(b_{f0} + T_b(t)\big)\, e^{-\alpha D}, \tag{12}$$

$$\dot{D} = [E - B]_+ - \eta D, \tag{13}$$

$$T_b(t) = \omega_{Tb}\, \frac{\Delta b}{1 + e^{-\kappa_w (t - t_{wave})}}, \qquad \omega_{Tb} \in \{0, 1\}. \tag{14}$$

The states and flows have distinct meanings and units. Areas A_f, A_c, and A_r are in hectares; B, E, and S are annual account amounts in gha, with the annual basis implicit. Debt D is a leaky model state driven by aggregate deficit $E - B$, measured in gha yr; both $\dot{D}$ and ηD are in gha. It is not an area, a financial liability, or an NFA-reported quantity.

The optional technology term T_b raises fast-land yield along a bounded logistic curve. It is off in the baseline, figures, and tables and appears only in the matched comparison in SI §S8. The dynamic states are A_f, A_c, q, P, and D. Timber liquidation and direct restoration from fast land to capital land are possible extensions, but neither is active in the reported runs (SI §S1).

Two feedbacks organize the model's behavior. First, when fast land contributes more annual biocapacity per hectare than capital land ($b_f > \Upsilon_c(q)$), conversion adds positively to B at the margin even while A_c falls. The net change in B also reflects maintenance, retirement, quality, and other active terms. A higher modeled B can raise delayed carrying capacity and, in an endogenous-demand run, later population and demand; this is a conditional feedback within the model, not a causal inference from NFA accounts.

Second, a shortfall adds to debt D, while repayment draws it down. As debt grows, b_f falls, which can reduce B and deepen the modeled shortfall. Under fixed demand, if the shortfall settles, debt approaches a finite level. It remains a model variable, not a measured physical quantity.

2.2. Parameter selection and empirical grounding

Table 1 summarizes the baseline values and evidence status. The model represents forests, soils, wetlands, and other slow-renewing systems with one effective capital-land class, despite their differences in area, condition, yield, and recovery. Parameter values are representative, not fitted; they illustrate mechanisms rather than forecast history.

Evidence from separate systems provides timescale context, not one pooled global rate: secondary-forest biomass reaches about half its reference level after a median of roughly 30 yr; the cited collapsed-fish-stock records give a median near 5.6 yr; and soil organic carbon approaches a new balance over about 17–23 yr (SI §S5.5) (Poorter et al., 2016; Hutchings and Reynolds, 2004; Neubauer et al., 2013; Poeplau et al., 2011). These observations are not interchangeable measures, and they do not estimate the model's lumped regeneration rate ρ_c or delay τ_g.

NFA cropland biocapacity per unit area and the separate cereal-yield/cropland-area proxy offer descriptive context only (SI §S5). They do not map one-to-one onto model fast land and are not fitted values of b_f or measurements of A_c. The baseline condition $b_f - \Upsilon_c(q) > 0$ is therefore an explicit scenario assumption: it defines when conversion contributes positively to the modeled total, not a universal empirical claim about farms and ecosystems.

2.3. Data and empirical measures

The empirical analysis uses the official National Footprint and Biocapacity Accounts (NFA), Edition 2026, results and metadata version 1.0 (Lo et al., 2026). We use complete world observations from 1961 through 2023 and exclude the estimated 2024–2025 rows. The 1961–2022 sensitivity uses the same account edition and population series, keeping the data vintage fixed.

Workbook measures. From the `world_data` sheet, we take total biocapacity B from `biocap_total_gha`, the Global Ecological Footprint of Production E_{prod} from `efp_total_gha`, and UN PopStat population from `pop_1000_persons`. We convert population from thousands to persons and calculate per-person values from unrounded totals rather than rounded workbook columns. Ratios, indices, and log changes are dimensionless.

Account classes and physical stocks. The workbook labels—Cropland, Grazing land, Forest biocapacity, Fishing grounds, and Built-up land—are NFA account values, not measures of biomass, soil condition, wetland extent, fish biomass, or model area A_c. Built-up land is an accounting class, not ecological capital. Changes in these rows describe account composition, not evidence of physical-stock trends or transfers among model land classes. SI §S5 gives variable definitions and endpoint calculations.

Table 1: Baseline parameter values and evidence status. Representative values are effective, lumped quantities; the NFA account/area quotient is descriptive, not a physical-yield measurement. Extension parameters are active only in scenarios specified in this table or the SI. Symbols and units are listed in SI §S1.

Parameter	Value	Meaning	Status
ρ_c	$0.08\,\mathrm{yr}^{-1}$	quality regeneration rate of capital land	representative; not fitted
b_{f0}	$0.85\,\mathrm{gha\,ha}^{-1}$	base fast-land yield per hectare	representative (2026 NFA account value/area quotient ≈ 1.045 in 1961; not a physical-yield measure)
b_c	$0.05\,\mathrm{gha\,ha}^{-1}$	capital-land direct yield	representative
$b_{G,c}$	$0.8\,\mathrm{gha\,yr\,ha}^{-1}$	regeneration-value coefficient for capital land	representative
$q_{\max}$	1.0 (dimensionless)	normalized quality ceiling	normalization
μ	$0.06\,\mathrm{yr}^{-1}$	cropland maintenance (reserve→fast)	representative
η_f	$0.05\,\mathrm{yr}^{-1}$	fast-land retirement (fast→reserve)	representative
τ_{conv}	1.0 yr	conversion time constant	representative
α	$0.03\,\mathrm{gha}^{-1}\,\mathrm{yr}^{-1}$	degradation of b_f per unit debt	representative
η	$0.05\,\mathrm{yr}^{-1}$	debt-repayment rate (> 0)	representative
e	$0.55\,\mathrm{gha\,cap}^{-1}$	per-capita footprint (constant)	representative
r	$0.02\,\mathrm{yr}^{-1}$	intrinsic population growth	representative
$\tau_g,\ \tau_p$	20, 25 yr	regeneration and demographic delays	representative; τ_p lumps institutional lags
A_{tot}	2.5 ha	total bioproductive area	normalization
$\sigma_f,\ \sigma_c$	1, 1 (dimensionless)	human-available shares of annual account amounts	baseline (< 1: reservation extension)
χ_r	$0.10\,\mathrm{gha}^{-1}\,\mathrm{yr}^{-1}$	surplus-gated restoration coefficient	on with endogenous demand; off under frozen demand
$A_c^{\min}$, $A_r^{\min}$, $K_{\min}$	0.05 ha, 0.10 ha, 0.02 cap	typed floors	domain constraints
Δb, κ_w, t_{wave}	$1.0\,\mathrm{gha\,ha}^{-1}$, $0.1\,\mathrm{yr}^{-1}$, 40 yr	technology wave	off in baseline; active only in the matched SI S8 experiment

Auxiliary cropland proxy. As a separate check, we pair OWID's FAOSTAT-derived global cropland-area series (arable land plus permanent crops) with cereal yield (Our World in Data, 2026a,b). Cereal yield is a limited productivity indicator, not a measure of all-crop output or an NFA yield factor. We scale the proxy to the NFA cropland share in 1961, calculate $X_t = \hat{b}_{f,t} A_{f,t}$, and define the remainder as $C_t = B_t - X_t$. This is an assumed account split: the residual absorbs differences in boundaries, categories, and calibration, and is not a measured physical stock or $\Upsilon_c(q)A_c$. SI §S5 reports the construction and calibration sensitivity. We infer no physical-stock trend from this proxy.

Total and per-person change. We use the exact identity $\Delta \ln B = \Delta \ln(B/P) + \Delta \ln P$ to separate change in the aggregate from change in biocapacity per person and population scale. It is an accounting identity, not a causal model of biocapacity production.

2.4. Numerical solution

We integrate the delay equations by the method of steps with explicit first-order Euler updates. The demonstration run and threshold bisection use $\Delta t = 0.05\,yr$; broader drawdown sweeps use $0.1\,yr$, with selected thresholds checked at $0.05\,yr$. The positive-part switch is exact. These checks document the resolution used, not convergence beyond tested settings.

We enforce $A_c \geq A_c^{\min}$, $A_f \geq 0$, $q \in [0, q_{\max}]$, $P, D \geq 0$, and $K \geq K_{\min}$. Histories of B and q remain at their initial values over the longest active delay. In the matched SI §S8 technology-wave experiment, both runs share that pre-run history and differ only in the logistic technology term; its small initial tail may affect current B before its later delayed effect on carrying capacity.

Floor-proximity bands reflect step-size and tolerance sensitivity, not sharp critical-population estimates. Primary runs use delayed quality.

3. Results

3.1. When conversion raises the total as capital land declines

For a given capital-land quality level q, its annual contribution per hectare, $\Upsilon_c(q)$, is a concave quadratic. It peaks at $q^* = q_{\max}/2$; with the baseline parameters, the maximum is $0.066\,\mathrm{gha\,ha^{-1}}$. The quality-related term contributes 0.016 of that value, while direct yield contributes $0.05\,\mathrm{gha\,ha^{-1}}$; the direct-yield component is larger in this case.

Conversion changes the aggregate according to the gap between the annual account contribution of one fast-land hectare and the effective value of one capital-land hectare. We call this gap the *composition diagnostic*:

$$\Delta b_{\text{conv}}(q, D) = b_f(D) - \Upsilon_c(q). \tag{15}$$

At baseline, Δb_{conv} equals +0.787, +0.784, and +0.794 gha ha^{-1} at $q = 0.3$, 0.6, and 0.9. Each lies in the composition-premium regime ($\Delta b_{\text{conv}} > 0$), where conversion adds to the aggregate at the margin. At $\Delta b_{\text{conv}} = 0$ the contribution is first-order neutral; below zero, it is a composition-discount. These labels describe the conversion term only. Maintenance, retirement, quality change, and other flows still determine the net change in B. At any given quality level, that contribution changes linearly with A_c; it has no interior maximum with respect to area.

Proposition 1 (Conversion, aggregate change, and what the total can reveal)**.** *Let $T_b \equiv 0$, $R_{rc} \equiv 0$ and hold D fixed, so that b_f is constant. Along any solution of (3)–(10),*

$$\dot{B} = \Delta b_{\text{conv}}(q)\, u_c + b_f\big(\mu A_r - \eta_f A_f\big) + b_{G,c}\, G_c'(q)\, A_c\, \dot{q}. \tag{16}$$

(i) During conversion ($u_c > 0$), A_c falls when restoration is off. The conversion component of $\dot{B}$ is $\Delta b_{\text{conv}}(q)u_c$, positive exactly when $\Delta b_{\text{conv}}(q) > 0$. It determines the sign of net $\dot{B}$ only if maintenance balances retirement ($\mu A_r = \eta_f A_f$) and quality is unchanged ($\dot{q} = 0$).

(ii) In the reduced comparison—with fixed demand and debt, baseline availability shares, balanced maintenance and retirement, and no quality change—a positive Δb_{conv} makes the shortfall shrink during conversion. The shortfall closes if the balance point is reached before the area floor binds. If the floor binds first, A_c approaches $A_c^{\min}$ and a shortfall may persist. Other active terms in the full model can change this result.

(iii) At fixed $b_f > 0$ and q, $B = b_f A_f + \Upsilon_c(q)A_c$ supplies one equation for the two land areas. Near an interior state, a small change in A_c can be offset by $\delta A_f = -(\Upsilon_c/b_f)\delta A_c$ and $\delta A_r = -\delta A_f - \delta A_c$, preserving both B and total area as long as the land bounds are respected. Thus several nearby allocations can have the same B; the aggregate alone does not identify A_c. For a finite observed change ΔB, the offset is $\Delta A_f = (\Delta B - \Upsilon_c \Delta A_c)/b_f$. It is feasible only if all resulting areas remain within

bounds, so the feasible range depends on the starting state. If A_f and b_f are observed independently, the residual $C = B - b_f A_f = \Upsilon_c(q)A_c$ identifies their product, not its two factors. An independent observation of A_c would give $\Upsilon_c(q) = C/A_c$. A single value of Υ_c cannot separate b_c, $b_{G,c}$, ρ_c, and q. Even with informative observations of q, it identifies at most b_c and the product $b_{G,c}\rho_c$, not $b_{G,c}$ and ρ_c separately.

Proof. Differentiate (3) with b_f constant and substitute (6)–(7) with $R_{rc} = 0$: $\dot{B} = b_f(u_c + \mu A_r - \eta_f A_f) - \Upsilon_c u_c + b_{G,c}G_c'(q)A_c\dot{q}$, which is (16). (i) Follows from (16) with $R_{rc} = 0$; when the maintenance–retirement imbalance and quality-growth terms vanish, only the conversion term remains. (ii) Under the reduced-case conditions in the statement, $S = E - B$ and $\dot{S} = -\dot{B} = -\Delta b_{\text{conv}} u_c < 0$ on $S > 0$. Since S is bounded below by zero and A_c is bounded below by $A_c^{\min}$, conversion either closes the shortfall or is limited by the floor. (iii) At fixed b_f and q, the observation map is linear; an interior feasible point admits the stated small perturbations along a level set of B, while finite changes must also obey the area bounds. The residual C identifies only $\Upsilon_c(q)A_c$ unless A_c is independently observed; the parameter combination follows from the definition of $\Upsilon_c(q)$. □

Figure 1a follows the baseline run with demand fixed at $E = 0.825$ gha and restoration off. Initial states are $A_f = 0.6$ ha, $A_c = 1.5$ ha, $A_r = 0.4$ ha, and $q = 0.9$. During conversion, model biocapacity B rises from 0.594 to 0.825 gha (39%); capital-land area falls from 1.500 to 0.808 ha (46%), while fast-land area grows from 0.600 to 0.923 ha.

Quality q rises from 0.9 to 1.0, moving farther above $q^* = 0.5$, where $\Upsilon_c(q)$ peaks. Its per-hectare value therefore falls from 0.05576 to 0.05000 gha ha^{-1}. The modeled capital-land contribution $\Upsilon_c(q)A_c$ falls from 0.08364 to 0.04038 gha (51.7%)—more than the 46.2% area decline because value per hectare also decreases. The one-to-one transfers conserve area: $\Delta A_f + \Delta A_c + \Delta A_r = 0$.

The adjustment is fastest early and then slows. On the deficit side, the approach to balance is exponential, with slow rate $|\lambda_1| = 0.054\,\text{yr}^{-1}$ and an e-folding time of about 19 yr. By year 25, 98% of the rise in B and 85% of the decline in A_c have occurred ($B = 0.820$ gha, $A_c = 0.913$ ha); by year 50, 97% of the area decline is complete. At the reported numerical resolution, $\dot{B} > 0$ and $\dot{A}_c < 0$ together for about 149 yr.

Reserve adjustment creates the long tail. As fast land retires to reserve, its contribution falls and a small shortfall reappears. Conversion then continues

until $A_r/A_f = \eta_f/\mu$. The duration is not a fixed horizon: the slow rate λ_1 sets its timescale, while Δb_{conv} and the initial shortfall affect its size (SI §S2.2).

The aggregate has a clear identification limit: B alone cannot show whether A_c is shrinking while another component grows. Independent observations of fast-land area A_f and yield b_f would let us calculate $C = B - b_f A_f$. But C identifies only the capital-land contribution, not its area and per-hectare value separately.

We call a trajectory with rising B and falling A_c a *composition mask*. A related but distinct pattern is the *liquidation mask*, in which demand E stays fixed while A_c falls. The first compares aggregate biocapacity with land area; the second compares demand with land area. They can occur together, but neither implies the other. The diagnostic Δb_{conv} gives the sign of conversion's marginal contribution, not necessarily the net change in B when other terms are active; a liquidation mask does not require $\Delta b_{\text{conv}} > 0$. Observing B, E, and A_c distinguishes their defining patterns. Here, "liquidation mask" names a trajectory; it is not the optional timber-liquidation extension, which is off in every reported run.

3.2. Fixed demand leads to a minimum-area boundary, not a fold

Once population reaches carrying capacity, $P = K = B/e$ and demand $E = eP$ equals biocapacity B; the shortfall then closes. Conversion is therefore transient in the endogenous-population runs. To isolate how the balance point changes with demand, we also fix E and analyze a reduced land system. The constraint $A_c \geq A_c^{\min}$ is a *typed floor*: it protects one land class, not the aggregate.

Proposition 2 (Balance point, demand ceiling and affine deficit-side dynamics)**.** *For this reduced comparison, hold demand at E, switch off the technology boost and restoration, set debt $D = 0$, and fix quality at $q = q_{\max}$. This quality is the attractor of (10) for $q(0) > 0$. Let $k = \mu/(\mu + \eta_f)$ and*

$$B_{eq}(A_c) = b_{f0}\, k\, (A_{tot} - A_c) + b_c A_c = b_{f0} k A_{tot} - (b_{f0} k - b_c) A_c. \tag{17}$$

(i) With no conversion ($S = 0$), reserve-to-fast maintenance balances fast-land retirement. This gives $A_f = k(A_{tot} - A_c)$ and $A_r = (1-k)(A_{tot} - A_c)$, where $k = \mu/(\mu+\eta_f)$. The resulting total is $B_{eq}(A_c)$. If $b_{f0} k > b_c$, this total decreases as A_c increases, so there is at most one balance point $B_{eq}(A_c^*) = E$:

$$A_c^*(E) = \frac{b_{f0} k A_{tot} - E}{b_{f0} k - b_c}. \tag{18}$$

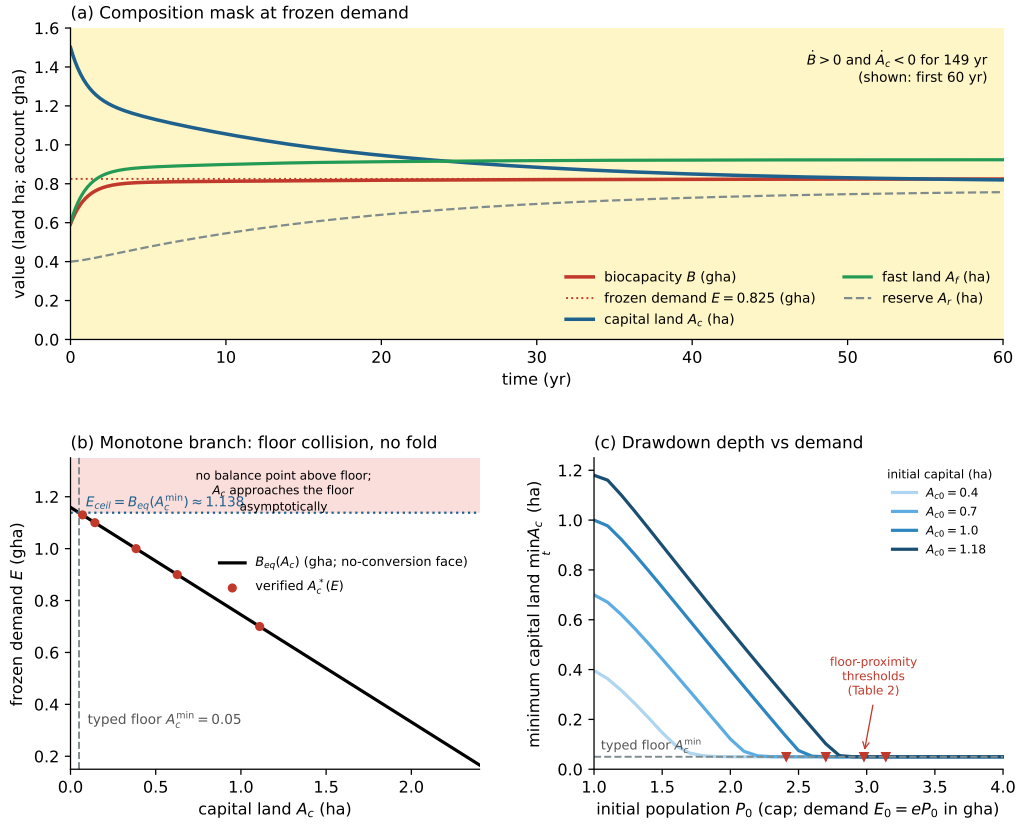


Figure 1: Baseline model results (Table 1). Account amounts B and E are in gha, land areas are in ha, and population is in cap. (a) Frozen demand ($E = 0.825$ gha), restoration off, and initial conditions $A_{f0} = 0.6$ ha, $A_{c0} = 1.5$ ha, and $q_0 = 0.9$. B rises from 0.594 to 0.825 gha as A_c falls from 1.500 to 0.808 ha and A_f rises from 0.600 to 0.923 ha. Shading marks the interval with $\dot{B} > 0$ and $\dot{A}_c < 0$ (about 149 yr; only the first 60 yr are shown). Within 25 yr, 98% of the rise in B and 85% of the fall in A_c occur. (b) Fixed-demand balance relation $B_{eq}(A_c)$; the listed balance areas are $A_c^*(E) = 1.110$, 0.626, 0.385, 0.143, and 0.070 ha at $E = 0.70$, 0.90, 1.00, 1.10, and 1.13 gha. The curve meets the minimum $A_c = 0.05$ ha at $E_{\text{ceil}} \approx 1.138$ gha; no balance point is admissible at higher demand. (c) Endogenous-demand runs with $\tau_g = 20$ yr, $\tau_p = 25$ yr, and $\chi_r = 0.10\,\text{gha}^{-1}\,\text{yr}^{-1}$. For each initial area A_{c0}, the minimum area reached varies continuously and monotonically with P_0 ($E_0 = eP_0$). The floor is approached asymptotically. Triangles mark the proximity values in Table 2, defined as within 10^{-6} ha of the floor; no discontinuity is observed.

The algebraic solution is feasible when $B_{eq}(A_{tot}) \le E \le E_{\text{ceil}}$, where $E_{\text{ceil}} = B_{eq}(A_c^{\min})$. It lies above the minimum area exactly when $E < E_{\text{ceil}}$. At lower demand, the algebraic root lies above A_{tot}, and every feasible no-conversion allocation already supplies at least E.

(ii) While shortfall is positive and conversion is not yet limited by the minimum-area constraint, the land subsystem is affine in $z = (A_f, A_c)$: $\dot{z} = Jz + c$, with

$$J = \begin{pmatrix} -\tau_{\text{conv}}^{-1} - \mu - \eta_f & -\mu - \dfrac{b_c}{b_f \tau_{\text{conv}}} \\ \tau_{\text{conv}}^{-1} & \dfrac{b_c}{b_f \tau_{\text{conv}}} \end{pmatrix}, \qquad \det J = \frac{\mu + \eta_f}{\tau_{\text{conv}}} \Big(k - \frac{b_c}{b_f} \Big), \tag{19}$$

and $\operatorname{tr} J < 0$. The matrix does not depend on demand or on the current state. Its determinant is positive exactly when B_{eq} decreases, and its fixed point is the balance point in (i). For $B_{eq}(A_{tot}) \le E < E_{\text{ceil}}$, trajectories that remain in this regime approach that point at rates set by the eigenvalues of J. For $E > E_{\text{ceil}}$, the balance point would lie below the allowed minimum: A_c approaches $A_c^{\min}$ asymptotically and the shortfall tends to $S^* = E - E_{\text{ceil}} > 0$. Thus the reduced balance curve meets the minimum-area boundary; it does not turn back or form a fold. The leading eigenvalue does not approach zero at $E = E_{\text{ceil}}$, and the reduced equilibrium relation has no hysteresis. These conclusions hold under the stated assumptions and do not imply that the realized land allocation is independent of its history.

(iii) Above the demand ceiling, with degradation active, debt approaches a unique value $D^*(E)$ satisfying $\eta D = E - B_{eq}(A_c^{\min}; D)$, where $B_{eq}(\cdot; D)$ uses yield $b_{f0} e^{-\alpha D}$. This result requires $\eta > \alpha b_{f0} k (A_{tot} - A_c^{\min})$; the equilibrium debt increases with demand.

Proof. (i) With $S = 0$ and $R_{rc} = 0$, (7) gives $\mu A_r = \eta_f A_f$, hence $A_r / A_f = \eta_f / \mu$ and, with $A_r = A_{tot} - A_f - A_c$, $A_f = k(A_{tot} - A_c)$; substituting into (3) with $q = q_{\max}$ ($\Upsilon_c = b_c$) and $D = 0$ gives $B = B_{eq}(A_c)$, and $S = 0$ requires $B \ge E$. Strict monotonicity and (18) follow from $b_{f0} k > b_c$. (ii) On the deficit side $u_c = (E - b_f A_f - b_c A_c)/(b_f \tau_{\text{conv}})$ is affine in z; substituting into (6)–(7) with $A_r = A_{tot} - A_f - A_c$ gives J and a constant vector c by direct differentiation; $\det J$ follows by expansion, and $\operatorname{tr} J = -\tau_{\text{conv}}^{-1}(1 - b_c/b_f) -$

$\mu - \eta_f < 0$ for $b_c < b_f$. The fixed point solves $u_c = 0$ and $\mu A_r = \eta_f A_f$, i.e. it is the balance point. Because J is constant, convergence rates are constant along the branch. (iii) The map $D \mapsto \eta D + B_{eq}(A_c^{\min}; D) - E$ is continuous, negative at $D = 0$ when $E > E_{\text{ceil}}$, tends to $+\infty$, and is strictly increasing when $\eta > \alpha b_{f0} k (A_{tot} - A_c^{\min}) \geq -\partial_D B_{eq}$; hence it has a unique root, which increases with E. □

The uniqueness result concerns the reduced equilibrium relation; it does not make realized trajectories independent of their history. In the baseline, $R_{rc} = 0$ and conversion has no reverse flow. Once the shortfall closes, A_c stops falling and approaches its lower balance value. It remains below its pre-overshoot level because restoration is disabled. Meanwhile, q evolves separately, so $\Upsilon_c(q)$ can still change as A_c approaches that value. The reduced system therefore has a unique frozen-demand balance point under Proposition 2's assumptions, alongside a history-dependent post-overshoot allocation. This is not perpetual decline at equilibrium and is not a claim about real ecosystems.

At baseline, $k = 0.5455$, the balance-branch slope parameter is $b_{f0}k - b_c = 0.414\,\text{gha}\,\text{ha}^{-1}$, and the demand ceiling is $E_{\text{ceil}} = 1.138\,\text{gha}$. Equation (18) predicts balance areas of 1.110, 0.626, 0.385, 0.143, and 0.070 ha at demands of 0.70, 0.90, 1.00, 1.10, and 1.13 gha; these match the integrated equilibria (Fig. 1b). The eigenvalues of J are real and negative ($\lambda_1 = -0.0537$, $\lambda_2 = -0.998\,\text{yr}^{-1}$). Across the tested fast-land yields, λ_1 shifts only from -0.0522 at $b_f = 0.60$ to $-0.0556\,\text{yr}^{-1}$ at $b_f = 2.00$ and never changes sign (Table 2a).

Above the ceiling, the calculated debt equilibria are $D^* = 1.19$ and $5.38\,\text{gha}\,\text{yr}$ at $E = 1.158$ and $1.238\,\text{gha}$; numerical integration gives 1.21 and $5.39\,\text{gha}\,\text{yr}$. The uniqueness condition holds with margin: $\alpha b_{f0} k (A_{tot} - A_c^{\min}) = 0.034\,\text{yr}^{-1}$, below $\eta = 0.05\,\text{yr}^{-1}$. Under fixed demand, debt therefore saturates rather than growing without bound.

Critical slowing down refers to a fold approach in which the slowest return rate tends to zero and the return time $T_{ret} \approx -1/\lambda_1$ grows. That does not occur along this reduced balance branch: its linearization is constant, λ_1 stays away from zero, and the return time remains near 19 yr as E approaches E_{ceil}. The modeled signal is instead A_c approaching its minimum—information an aggregate-only observation cannot provide.

When population responds to supply, the minimum A_c reached varies continuously and monotonically with initial population P_0 for each starting area A_{c0} (Fig. 1c). The trajectory approaches, but does not cross, the floor

because $u_c \leq (A_c - A_c^{\min})/\tau_{\text{conv}}$. A larger initial capital-land area supports a larger P_0 before the run approaches that floor.

Table 2b gives the initial populations for which A_c comes within 10^{-2}, 10^{-4}, or 10^{-6} ha of its minimum. These tolerance-based values are proximity markers, not abrupt thresholds. At the tightest tolerance, $P_0 \approx 2.41$, 2.70, 2.98, and 3.14 cap for $A_{c0} = 0.40$, 0.70, 1.00, and 1.18 ha, respectively. They rise with starting area; no jump appears at any tested tolerance.

At $A_{c0} = 1.0$ ha, the 10^{-6} ha proximity value shifts by at most 0.001 cap across $\tau_g = 5$, 20, and 40 yr (SI Table S2). This small change applies only to the tested range; it does not show that delay is irrelevant more generally. The threshold is also unchanged to three decimal places when surplus-gated restoration is switched on (SI §S4), because that flow is inactive while $B < E$.

Table 2: Model results at the baseline parameter set. (a) At $q^* = q_{\max}/2$ ($\Upsilon_c(q^*) = 0.066$ gha ha^{-1}), the table reports $\Delta b_{\text{conv}}(q^*)$, the eigenvalues of the deficit-side matrix J in (19), and the demand ceiling across a sweep of b_f; all other parameters are at baseline. (b) For endogenous-demand runs with $A_{f0} = 0.6$ ha, $q_0 = 0.9$, $\tau_g = 20$ yr, and $\tau_p = 25$ yr, the table reports the initial population P_0 at which the minimum A_c comes within the stated distance of the floor $A_c^{\min} = 0.05$ ha. The first two columns use a P_0 grid with step 0.1 ($\Delta t = 0.1$ yr, $T = 400$ yr); the final column uses bisection with the 10^{-6} ha proximity criterion ($\Delta t = 0.05$ yr, $T = 1000$ yr). Because drawdown depth varies continuously with P_0 (Fig. 1c), these values define proximity bands, not a single threshold.

(a) Fast-land yield sweep (all rows composition-premium, $\Delta b_{\text{conv}} > 0$)

b_f (gha ha^{-1})	$\Delta b_{\text{conv}}(q^*)$ (gha ha^{-1})	λ_1 (yr^{-1})	λ_2 (yr^{-1})	E_{ceil} (gha)
0.60	+0.534	−0.0522	−0.975	0.804
0.85 (baseline)	+0.784	−0.0537	−0.998	1.138
1.20	+1.134	−0.0547	−1.014	1.606
2.00	+1.934	−0.0556	−1.029	2.675

(b) Initial population P_0 (cap) at which $\min_t A_c$ approaches $A_c^{\min}$ to within:

A_{c0} (ha)	within 10^{-2} ha	within 10^{-4} ha	within 10^{-6} ha
0.40	≈ 1.8	≈ 2.1	≈ 2.41
0.70	≈ 2.2	≈ 2.4	≈ 2.70
1.00	≈ 2.6	≈ 2.8	≈ 2.98
1.18	≈ 2.8	≈ 3.0	≈ 3.14

3.3. Technology gains are finite; debt depends on the shortfall

The technology boost and debt both change fast-land yield b_f, but in different ways. The boost has a fixed ceiling. Debt can accumulate while a shortfall persists, then approaches a finite level if that shortfall settles.

Proposition 3 (The technology boost is capped; debt depends on the shortfall)**.** *Under* (12)–(14)*:*

(i) The technology factor $(b_{f0}+T_b(t))/b_{f0}$ *stays between* 1 *and* $1+\Delta b/b_{f0}$*. Its log-scale gain is therefore bounded by* $\ln(1+\Delta b/b_{f0})$ *for all time (*0.78 *at baseline).*

(ii) Debt satisfies $D(t) \le \max\{D(0),\ \sup_{s\le t}[E-B]_+(s)/\eta\}$*. If the shortfall settles at* S^**, then* D *tends to* S^*/η*. The resulting reduction in fast-land yield depends on the size of that sustained shortfall.*

(iii) The debt-related yield loss exceeds the largest possible technology gain once the sustained shortfall exceeds $S^\dagger = (\eta/\alpha)\ln(1+\Delta b/b_{f0})$*, which is* 1.30 *gha at baseline.*

With additive degradation, $b_f = b_{f0}+T_b-\alpha D$*, the two effects enter symmetrically; the comparison in (iii) then reduces to* Δb *versus* αD^**. The asymmetry above follows from the model's multiplicative form.*

Proof. (i) is $0 \le T_b \le \Delta b$. (ii) From (13), $\dot{D} \le \bar{S} - \eta D$ with $\bar{S} = \sup_{s\le t}[E-B]_+$, so $D(t) \le \max\{D(0), \bar{S}/\eta\}$ by comparison; if $[E-B]_+ \to S^*$ then $D \to S^*/\eta$. (iii) compares $\alpha S^*/\eta$ with $\ln(1+\Delta b/b_{f0})$. □

With fixed demand and a persistent shortfall, D approaches S^*/η; larger shortfalls imply higher debt. At baseline, the crossover $S^\dagger \approx 1.3$ gha exceeds baseline B. Under this parameterization, the minimum-area constraint limits conversion before debt-related yield loss exceeds the largest possible technology gain. Debt nevertheless reduces fast-land yield over time.

A technology boost can raise B and carrying capacity; delayed population growth may raise demand too. Peak debt therefore depends on initial population and the summary window. In the matched SI §S8 experiment, for $P_0 = 1.0$ cap, peak debt over 0–250 yr is 0.02020 gha yr with the boost and 0.00813 gha yr without it (ratio 2.49). For $P_0 = 1.2$ cap, the full-run maximum is lower with the boost (0.05104 versus 0.06122 gha yr), but the maximum after the 40 yr wave midpoint is higher (0.02518 versus 0.01085 gha yr; ratio 2.32). These are conditional model results, not forecasts or empirical estimates.

3.4. Recovery requires an explicit, gated restoration flow

Conversion moves land from capital to fast use. The model has no route back to capital land unless restoration R_{rc} is active; fast-land retirement returns land to reserve instead. Lowering population, footprint, or shortfall therefore does not, by itself, reverse earlier conversion. With $\chi_r = 0$, A_c stays at its post-conversion level.

In one endogenous-demand run ($A_{c0} = 1.0\,\mathrm{ha}$, $P_0 = 2.0\,\mathrm{cap}$), A_c drops to 0.398 ha within four years and remains there without restoration. With surplus-gated restoration ($\chi_r = 0.10\,\mathrm{gha^{-1}\,yr^{-1}}$), it recovers to 0.828 ha, a recovery fraction of $\mathcal{R}_c = (A_c(T) - A_c^{\mathrm{deg}})/(A_c^{\mathrm{init}} - A_c^{\mathrm{deg}}) = 0.71$ (SI §S4.3).

The two restoration gates operate under different conditions. Surplus-gated restoration is inactive during a shortfall, but may continue after a surplus opens while population adjusts. Reaching a specified endpoint therefore requires a target-limited rule (SI §S4.4). Deficit-gated restoration acts during shortfall, when supply is already insufficient. At the tested rates, conversion outpaces restoration and drawdown continues. The result depends on timing and rates; neither gate is universally preferable.

3.5. NFA account composition, 1961–2023

For the primary empirical analysis, we use complete world observations in NFA Edition 2026, results and metadata version 1.0, and exclude the estimated 2024–2025 values. Our 1961–2022 sensitivity uses the same account edition and population series.

Between 1961 and 2023, total biocapacity account value B rises from 9.7254×10^9 to 12.1428×10^9 gha (24.86%; $\Delta \ln B = +0.221997$). Population grows from 3.06487 to 8.09173 billion. Over the same interval, B/P falls from 3.17319 to 1.50065 $\mathrm{gha\,person^{-1}}$ (52.71%; $\Delta \ln(B/P) = -0.748840$). The Global Ecological Footprint of Production per person (E_{prod}/P) rises from 2.19700 to 2.56462 $\mathrm{gha\,person^{-1}}$ (16.73%). Meanwhile, E_{prod}/B rises from 0.69236 to 1.70901, first exceeding one in 1973.

The exact identity $\Delta \ln B = \Delta \ln P + \Delta \ln(B/P)$ yields $+0.970837 - 0.748840 = +0.221997$. It separates population-scale change from change in biocapacity per person, but does not explain their causes. The same-edition 1961–2022 sensitivity gives $\Delta \ln B = +0.219497$, a 52.41% fall in B/P, a 17.70% rise in E_{prod}/P, and 1973 as the first year with $E_{\mathrm{prod}}/B > 1$ (SI §S5).

The component series show how NFA account composition changed (Fig. 3; Table 3). Cropland contributes +24.06 percentage points to the aggregate's +24.86% change—about 96.8% of the net percentage change.

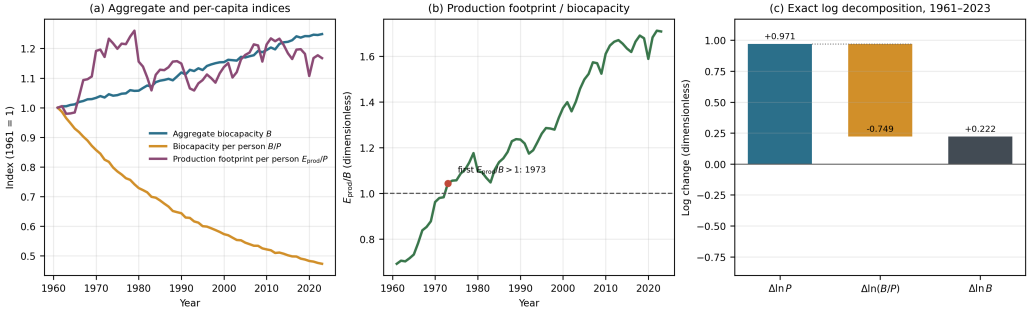


Figure 2: Aggregate and per-capita movements in NFA Edition 2026 (world, 1961–2023; complete observations). B is total biocapacity, E_{prod} is the workbook's Global Ecological Footprint of Production, and P is the single UN PopStat population series; per-capita values are calculated from totals. (a) Indices (1961 = 1) of B, B/P, and E_{prod}/P. (b) E_{prod}/B, first above one in 1973. (c) Exact log identity $\Delta \ln B = \Delta \ln P + \Delta \ln(B/P)$. Account values are in gha; plotted indices, ratios, and log changes are dimensionless. The 2024–2025 estimated rows are excluded.

Forest biocapacity contributes −5.95 points, grazing land −0.27, fishing grounds −0.42, and built-up land +7.44. The non-cropland account sum, which includes built-up land, is nearly unchanged (1.0092×, or +0.80 points) because the built-up increase offsets declines elsewhere. These are changes in account values. They do not establish changes in forest biomass, soil condition, wetland extent, fish biomass, or model area A_c. Built-up land is an account class, not ecological capital. Component values sum to total biocapacity within workbook rounding at both endpoints.

Table 3: NFA Edition 2026 biocapacity-account endpoint ratios and contributions. Contributions are $100\,[B_i(T) - B_i(1961)]/B(1961)$ percentage points; endpoint ratios are $B_i(T)/B_i(1961)$. The 2022 sensitivity is from the same edition. All entries are account values, not physical-stock measurements.

Component	2023 ratio	2023 contribution (pp of B_{1961})	2022 sensitivity ratio	2022 contribution (pp of B_{1961})
Cropland	2.8701	+24.0577	2.8514	+23.8183
Grazing land	0.9828	−0.2688	0.9815	−0.2890
Forest biocapacity	0.8996	−5.9525	0.9019	−5.8137
Fishing grounds	0.9627	−0.4212	0.9596	−0.4559
Built-up land†	8.7935	+7.4416	8.6298	+7.2854
Non-cropland sum†	1.0092	+0.7990	1.0083	+0.7267
Total biocapacity	1.2486	+24.8568	1.2454	+24.5450

†Built-up land is not ecological capital; the non-cropland account sum includes it. Component sums match total biocapacity within workbook rounding.

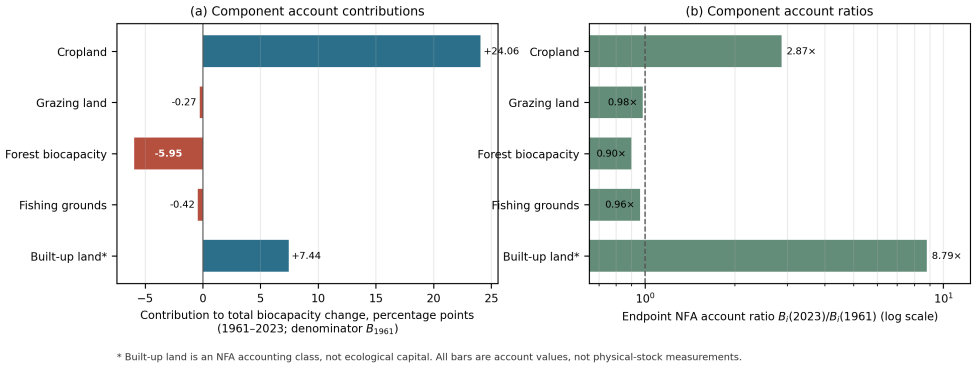


Figure 3: NFA Edition 2026 land-type biocapacity-account changes, world, 1961–2023. (a) Component contributions in percentage points of B_{1961}. (b) Endpoint ratios $B_i(2023)/B_i(1961)$. "Forest biocapacity" is the workbook's exact label. All bars are account values, not measurements of physical stocks; built-up land is an NFA accounting class, not ecological capital.

We also use external cropland area and cereal yield to build an illustrative proxy for the fast-land contribution (SI §S5). Anchoring it to the measured 1961 NFA cropland share ($\phi_0 = 0.128647$, about 12.86%), the exact log-change split assigns $+0.268662$ to the cereal-yield index, $+0.037710$ to cropland area, and -0.084375 to the residual. Together these sum to $\Delta \ln B = +0.221997$. The identity is exact for this proxy, but its allocation changes with the anchor and is not a physical-stock attribution.

3.6. Falsifiable predictions

The model yields five testable predictions. Testing its composition mechanism requires more than aggregate biocapacity: it requires information on capital-land area, fast-land area, and relevant land-cover or yield proxies.

P1. **Conversion can raise the total under a specific condition.** In the baseline frozen-demand run, $\dot{A}_c < 0$ while $\dot{B} > 0$ for a finite interval (Fig. 1a). Conversion adds to B exactly when $\Delta b_{\text{conv}} > 0$, although other flows, quality, debt, technology, and restoration determine the net change. With baseline shares $\sigma_f = \sigma_c = 1$, conversion requires demand to exceed total biocapacity ($R_B = E/B > 1$). The reserved-share extension can permit conversion when $R_B \leq 1$, but that is not a baseline result. A composition mask is rising B with falling A_c; a liquidation mask is steady E with falling A_c. The two patterns may overlap but are not interchangeable. The sign of Δb_{conv} describes conversion's contribution, not necessarily the net change in B.

P2. **The duration depends on model rates, not a universal horizon.** The slow return rate λ_1 (along with τ_{conv}, μ, η_f, and b_c/b_f) sets the timescale; Δb_{conv} and the initial shortfall affect the size of the change. In the baseline, most conversion occurs within one to two e-folding times, when the remaining gap shrinks by a factor of e.

P3. **The reduced model has no fold-type early warning.** Return time does not grow as demand approaches the ceiling, and drawdown depth changes continuously with demand (Fig. 1c). The model instead signals that A_c is approaching its minimum. It predicts no fold-related warning before that boundary.

P4. **The population proximity value depends on starting area and is weakly delay-sensitive in the tested range.** The initial population at which a run approaches the capital-land floor rises with starting A_c (Table 2b). For $A_{c0} = 1.0\,\text{ha}$, the $10^{-6}\,\text{ha}$ proximity value shifts by at most $0.001\,\text{cap}$ across $\tau_g = 5$, 20, and $40\,\text{yr}$ (SI Table S2). This comparison does not establish delay invariance beyond that range; the small shift is consistent with the modest regenerative contribution and monotone quality path in the baseline.

P5. **Recovery requires a return flow.** Lower demand alone does not restore capital-land area. The model needs an explicit restoration flow. A surplus-gated flow is inactive during shortfall, whereas a deficit-gated flow can further reduce provision.

Aggregate B alone cannot test these predictions. The composition mechanism requires independent observations or defensible proxies for A_c and fast-land yield b_f. Testing delay, recovery, and demand-threshold predictions also requires relevant time series and intervention or demographic data.

4. Discussion

4.1. What the two-land model shows

The two-land model yields three results relevant to human–environment models (Schaefer, 1954; May, 1973; Meadows et al., 1972, 2004; Wackernagel and Rees, 1996; Brander and Taylor, 1998). First, conversion can add to total biocapacity while A_c falls. Whether B rises overall also depends on other land flows and quality change; Proposition 1 gives the exact condition for the conversion term. The technology boost is capped, while debt depends on the magnitude and duration of shortfall (Proposition 3).

Second, under the reduced fixed-demand assumptions ($q = q_{\max}$, $D = 0$, technology boost off, and no restoration), the balance relation declines steadily with A_c until it meets the area floor. It does not fold back. The absence of a fold, hysteresis, or critical slowing applies to this equilibrium relation, not to every trajectory. With one-way conversion and no return flow, A_c remains below its pre-overshoot level after conversion stops, although it no longer declines at the balance point (Table 4). Third, recovery requires an explicit return flow; its effect depends on whether the restoration gate opens during a shortfall or a surplus.

SI §S7 gives a separate one-stock comparator. Its state A has comparator-specific units; it is not A_c. With constant parameters and no debt, its equilibrium relation is $B(A) = (b + b_G\rho)A - (b_G\rho/A_{\max})A^2$, with $P = B(A)/e$. At the illustrated settings, $b_G\rho < b$, so the relation is monotone; an interior maximum would require other parameters. Its neutral equilibrium family and selected delay trajectories belong only to that formulation. They establish neither a fold nor a universal critical delay or collapse mechanism in the two-land model.

These results also clarify when to expect common early-warning indicators. Rising autocorrelation and variance are associated with approaching a fold (Scheffer et al., 2001). This reduced model reaches a minimum-area boundary instead, so those indicators need not rise first. Their absence would not establish safety. The relevant model signal is the path of A_c, which the aggregate alone cannot reveal.

Table 4: Equilibrium, trajectory, and tested delay-sensitivity properties under the stated assumptions.

Property	Finding	Scope
Fold on the reduced frozen-demand branch	Absent	$B_{eq}(A_c)$ is strictly monotone; the balance point collides with the typed floor at $E_{\text{ceil}} \approx 1.138\,\text{gha}$ (Proposition 2)
Critical slowing as $E \to E_{\text{ceil}}$	Absent	The slow eigenvalue remains negative and constant along the reduced branch
Hysteresis of the frozen-demand equilibrium relation	Absent	There is at most one feasible balance point; this does not imply history independence of realized states
Path dependence of realized A_c	Present, conditional	With $R_{rc} = 0$, one-way conversion leaves A_c below its pre-overshoot level after shortfall closure, while $\dot{A}_c = 0$ at equilibrium
Discontinuous response to endogenous demand	Not observed	In the reported runs, minimum A_c varies continuously and monotonically with initial population; proximity thresholds depend on tolerance
Regeneration-delay sensitivity (τ_g, with $\tau_p = 25\,\text{yr}$)	Small in tested range	At $A_{c0} = 1.0\,\text{ha}$, the delayed-quality 10^{-6} ha threshold varies by 0.001 cap across $\tau_g = 5, 20, 40\,\text{yr}$ (SI Table S2); this does not establish invariance outside that range

4.2. Measurement implications

Biocapacity in gha is a value-weighted total, not an inventory of land classes. A 20% increase cannot distinguish modest gains across all components from a large gain in one that offsets losses elsewhere. Proposition 1(iii) formalizes the limit: even with fixed per-hectare yields, the total and the conserved-area constraint leave multiple feasible allocations among fast, capital, and reserve land.

A related transition-assessment study defines a linear compensatory operator and compares perfect substitutability with less-substitutable CES alternatives (Sections 2 and 4 of Abaee, 2026b). ECOMOD's B is also a linear sum, but the companion analyzes a different, finite-plan setting. There, the compensatory protocol accepts states even when no single available plan meets every component floor (Theorem 2 in Abaee (2026b)). Theorem 4(iii) in Abaee (2026b) identifies an exact interval of diagonal-gap states on its witness-defined region that each positive elasticity falsely certifies as safe; thus no positive-elasticity member is uniformly safe on that region. If feasibility is required throughout a transition, falling below a land-class minimum remains a violation even if the endpoint later recovers.

For the NFA, these findings concern account composition, not physical stocks. In Edition 2026, total world biocapacity account value B rises 24.86% from 1961 to 2023, while B/P falls 52.71%. The Global Ecological Footprint of Production per person (E_{prod}/P) rises 16.73%, and E_{prod}/B first exceeds one in 1973. B and E_{prod} are annual account amounts in gha, with the annual basis implicit; per-person values are in gha per person, and E_{prod}/B is dimensionless.

Cropland contributes +24.06 percentage points to the total change; Forest biocapacity contributes −5.95, grazing land −0.27, fishing grounds −0.42, and built-up land +7.44. The non-cropland sum is nearly flat because these account movements offset one another, including the large built-up-land increase. SI §S5 reports the same-edition 1961–2022 comparison. None of these values measures forest biomass or condition, soil carbon, wetland extent, fish biomass, or another physical stock. In particular, a decline in Forest biocapacity does not establish a decline in forest biomass. Built-up land is an NFA accounting class, not ecological capital. The ratios, contributions, vintage, and endpoint sensitivity describe account composition only (Lo et al., 2026; Borucke et al., 2013; Blomqvist et al., 2013; Giampietro and Saltelli, 2014; Galli et al., 2016; van den Bergh and Grazi, 2015).

The model result is distinct from the NFA trends. In the frozen-demand baseline (Fig. 1a), A_c falls from 1.500 to 0.807659 ha (46.16%), while model B rises from 0.594 to 0.825 gha (39%). Quality q rises from 0.9 to 1.0, moving farther above $q^* = q_{\max}/2$, where $\Upsilon_c(q)$ peaks. Its per-hectare value therefore falls from 0.05576 to 0.05000 gha ha^{-1} (10.33%), and the modeled contribution $\Upsilon_c(q)A_c$ falls from 0.08364 to 0.04038 gha (51.72%).

With restoration off, one-way conversion leaves A_c below its starting value after the shortfall closes. It then remains at the lower balance level rather than declining indefinitely. This is a conditional model result, not an observation or estimate of global physical-stock change.

We assembled no common-boundary global physical-stock series for these components and therefore report no physical-stock composite or global decline finding. The SI scoreboard is a specification—not an inventory—with candidate stock, extent, condition, and annual-flow measures and the requirements for comparing them; it contains no endpoint values. Until spatial and temporal boundaries, units, uncertainty, overlap, valuation or normalization, and permitted substitution are specified, these unlike measures should remain a nonadditive vector. SEEA Ecosystem Accounting also reports ecosystem extent and condition separately from ecosystem services and monetary presentations (United Nations et al., 2021).

The external cropland-area and cereal-yield series provide an illustrative proxy, not direct measurements of the model's land classes. The proxy is anchored to the 1961 NFA cropland share; cereal yield is a productivity sentinel, not an all-crops measure, and the area and account boundaries do not match by construction. External cropland area A_f is in hectares and index-scaled yield $\hat{b}_f$ in gha ha^{-1}, so $X = \hat{b}_f A_f$ and $C = B - X$ are account amounts in gha. The residual can absorb boundary and calibration differences, other account components, and conversion-factor changes. It is neither $\Upsilon_c(q)A_c$ nor a physical-stock measure. Physical interpretation would require independent, boundary-consistent measures of area, condition, yield, and regeneration, together with a validated mapping and valuation. Forest area could inform one part of the effective capital class, but cannot represent the entire class or its condition.

4.3. Policy controls and limits

We compare three controls. A conversion stop triggered at $A_c = A_c^{\min}$ preserves the capital-land floor. A ceiling on fast-land area A_f does not guarantee that floor unless the stop rule is explicitly tied to it and to the

remaining-area constraints. Nor can a heuristic governor that monitors only total B guarantee the floor: the aggregate does not identify A_c.

The restoration gates also behave differently. Surplus-gated restoration is inactive during a shortfall; deficit-gated restoration acts during the shortfall and can reduce provision further. This comparison echoes the distinction between protective and mobilizing institutional feedback in related work (Abaee, 2026a). With a common reserved share, $\sigma_f = \sigma_c = \sigma < 1$, conversion is inactive while $E \leq \sigma B$. With separate shares, the condition is $E \leq \sigma_f Y_f + \sigma_c Y_c$. This reservation applies to annual human-available flow, not to spatially protected area. Half-Earth, by contrast, is a spatial conservation target (Wilson, 2016). The connection is only the abstract idea of reservation: the model neither maps protected areas nor evaluates that proposal. Robustness to disturbances and parameter uncertainty remains untested.

4.4. Relation to prior work

The ledger companion distinguishes weak and strong sustainability as operating regimes according to whether material loops close at throughput. In its terms, a positive weighted sum ranks outcomes under weak sustainability, whereas componentwise adequacy is the strong-sustainability predicate (Abaee, 2026c, Sections 1.1 and 10.1). ECOMOD's B is an annual biocapacity-account total. The link is an analogy about compensation, not an identity with the companion's material-flow ledger or a complete sustainability classification. The assessment companion studies aggregate measures and typed transition criteria in a separate finite-plan setting (Abaee, 2026b); the governance companion studies institutional feedback (Abaee, 2026a). The delays here represent ecological regeneration and demographic response, not institutional review.

4.5. Limitations

Several limitations set the scope of the results. Ecological-capital land combines unlike assets, so its single quality state, regeneration rule, composition diagnostic, and demand ceiling are effective model quantities—not estimates for forests, soils, or wetlands. Most parameters are representative rather than fitted. A cross-section cannot identify b_c, $b_{G,c}$, and ρ_c separately. Recovery records from different systems provide timescale context, not a global estimate of ρ_c; they describe regeneration responses, not the delay τ_g. We hold the per-person footprint constant to isolate feedbacks among land,

biocapacity, and demand. A changing footprint would require another model extension.

The composition pattern depends on demand and on the mechanisms included here; under the stated assumptions, it is bounded. The exact shortfall switch makes stability at $S = 0$ one-sided. Drawdown thresholds are reported as proximity bands because they depend on tolerance and time step. We fix debt repayment at $\eta = 0.05\,\mathrm{yr}^{-1}$; the zero-repayment limit ($\eta \to 0$), which removes the debt equilibrium, is not analyzed. NFA accounts combine area and yield and therefore cannot identify them separately. Finally, replacing delayed regeneration $G_c(q(t-\tau_g))$ with current-quality regeneration $G_c(q(t))$ lengthens the demonstration mask by 1.3 yr and shifts the initial-population threshold by at most 0.001 cap (SI §S3).

5. Conclusions

An aggregate biocapacity total cannot reveal how its land classes or their areas have changed. In the two-land model, a shortfall-triggered transfer adds positively to total model biocapacity when fast-land yield exceeds capital land's effective per-hectare value, even while A_c falls; other active terms determine the net change in B. In the representative run, A_c falls 46.16%, $\Upsilon_c(q)$ falls 10.33%, and its modeled contribution $\Upsilon_c(q)A_c$ falls 51.72%, while total model B rises 39%. These are conditional model results, not measurements of global physical stocks.

The fixed-demand comparison follows a reduced balance branch to a typed area floor at a demand ceiling of about 1.14 gha, without a fold, hysteresis, or critical slowing on that branch. This does not establish the behavior of every trajectory in the delayed model. The realized path remains history-dependent; when restoration is off, capital-land area stays below its pre-overshoot level after the shortfall closes, although it no longer declines at the balance point.

NFA Edition 2026 reports a separate kind of evidence: account values. From 1961 to 2023, total world biocapacity account value rises 24.86%, while B/P falls 52.71% and the production footprint per person rises 16.73%. The component accounts show the composition of that change:

- Cropland contributes +24.06 percentage points to the total change.
- Forest biocapacity contributes −5.95 points.

- Built-up land contributes +7.44 points as an NFA accounting class, not ecological capital.

These movements do not measure biomass, soil condition, wetland extent, fish biomass, or the model's A_c, and they do not show that one land class caused another's change. The auxiliary crop proxy is an assumed account split, not a physical-stock measure. No common-boundary global physical-stock series was assembled; the SI scoreboard specifies candidate measures and comparison requirements, not an inventory. The 1961–2022 sensitivity uses the same NFA edition and population series.

The practical lesson is to read component accounts alongside the aggregate. The orchard is an analogy for the information problem, not evidence of a measured global stock decline: a harvest total is not a census of the orchard.

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Declarations

Data availability. All empirical inputs are publicly available from the sources cited in the manuscript. Derived analysis data and model code will be deposited in a public GitHub repository and archived in Zenodo; repository and record details will be added after the deposit is complete.

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